

MTH 203 : Lecture 04

Limit

Recall

Let $f: \mathbb{R} \rightarrow \mathbb{R}$

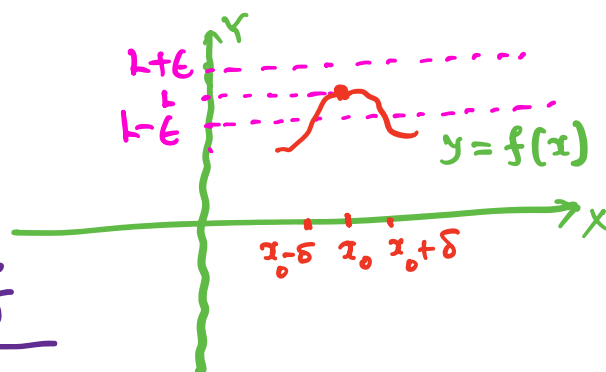
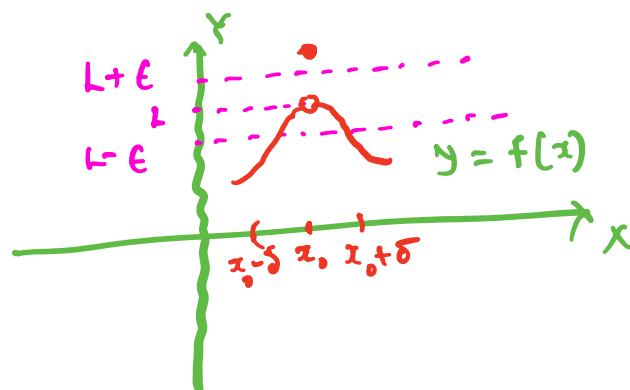
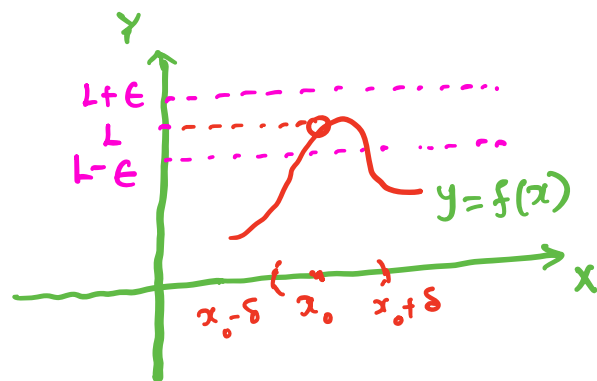
$\lim_{x \rightarrow x_0} f(x) \Leftrightarrow$ We can get $f(x)$ as close to L as desired (i.e. arbitrary close to L) by choosing x sufficiently close (but not equal to) x_0 .

$\Rightarrow$ given any $\epsilon > 0$,
(there exists) $\exists \delta = \delta(\epsilon) > 0$
such that
for all x in the domain of f

with $0 < |x - x_0| < \delta$

$\Rightarrow |f(x) - L| < \epsilon$

(i.e. $x_0 - \delta < x < x_0 + \delta \Rightarrow L - \epsilon < f(x) < L + \epsilon$)



Note:

• $x_0 \in \mathbb{R}$ but need not be in the domain of f .

• Even if $x_0 \in \text{Domain}(f)$, it may be the case that $f(x_0) \neq \lim_{x \rightarrow x_0} f(x)$.

• If $\lim_{x \rightarrow x_0} f(x) = f(x_0)$, we call f to be continuous at x_0 .

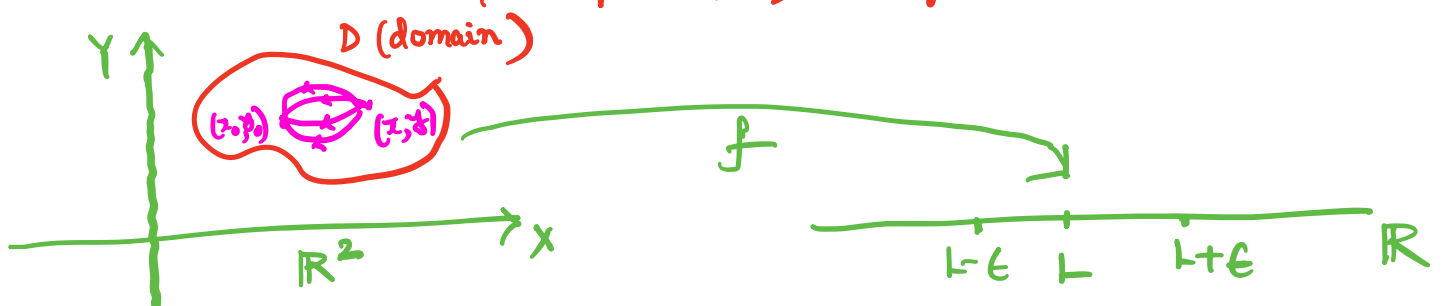
Now: Let $D \subseteq \mathbb{R}^2$ and $f: D \rightarrow \mathbb{R}$

We say that $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L$

$\Leftrightarrow$ given any $\epsilon > 0$, there exists a corresponding $\delta = \delta(\epsilon) > 0$ such that for all (x,y) in the domain of f ,

$$0 < \|(x,y) - (x_0,y_0)\| = \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta$$

$$\Rightarrow |f(x,y) - L| < \epsilon$$



Ex ① $f(x, y) = x$

$$\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = x_0$$

Ex ②: $f(x, y) = c$

$$\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = c$$

Ex ③ Find $\lim_{\substack{(x, y) \rightarrow (4, 3) \\ x \neq y+1}} \frac{\sqrt{x} - \sqrt{y+1}}{x-y-1}$ if it exists.

$$\lim_{\substack{(x, y) \rightarrow (4, 3) \\ x \neq y+1}} \frac{\sqrt{x} - \sqrt{y+1}}{x-y-1} = \lim_{\substack{(x, y) \rightarrow (4, 3) \\ x \neq y+1}} \frac{(\sqrt{x} - \sqrt{y+1})(\sqrt{x} + \sqrt{y+1})}{(x-y-1)(\sqrt{x} + \sqrt{y+1})}$$

$$= \lim_{\substack{(x, y) \rightarrow (4, 3) \\ x \neq y+1}} \frac{\cancel{(x-y-1)}}{\cancel{(x-y-1)}(\sqrt{x} + \sqrt{y+1})}$$

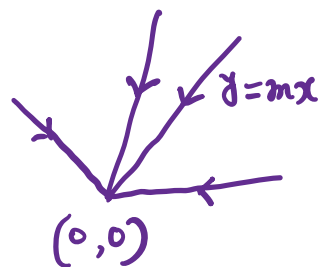
$$= \lim_{\substack{(x, y) \rightarrow (4, 3) \\ x \neq y+1}} \frac{1}{(\sqrt{x} + \sqrt{y+1})} = \frac{1}{\sqrt{4} + \sqrt{4}} = \boxed{\frac{1}{4}}$$

Ex ④: Let $g(x, y) = \frac{x-y}{x+y}$

Find $\lim_{(x, y) \rightarrow (0, 0)} g(x, y)$ if it exists.

Along the line $y = mx$,

$$g(x, y) = \frac{x - mx}{x + mx} = \frac{1 - m}{1 + m}$$



So, $\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } y=mx}} f(x,y) = \frac{1-m}{1+m}$ (different from different m)

For $m = \frac{1}{2}$, it is $\frac{\frac{1}{2}}{\frac{3}{2}} = \frac{1}{3}$
 For $m = 1$, it is 0
 For $m = 2$, it is $-\frac{1}{3}$

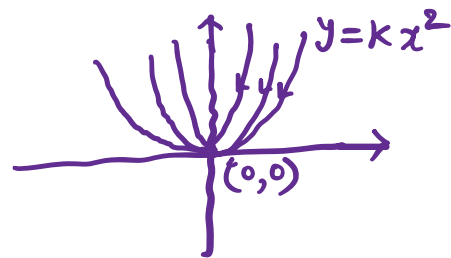
So, $\lim_{(x,y) \rightarrow (0,0)} \frac{x-y}{x+y}$ doesn't exist.

Ex 5: Let $f(x,y) = \frac{x^4 - y^2}{x^4 + y^2}$

Find $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ if it exists.

Along the parabola $y = kx^2$,

$$f(x,y) = \frac{x^4 - k^2 x^4}{x^4 + k^2 x^4} = \frac{1 - k^2}{1 + k^2}$$



So, $\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along } y=kx^2}} f(x,y) = \frac{1-k^2}{1+k^2}$, different for different k .

Thus $\lim_{(x,y) \rightarrow (0,0)} \frac{x^4 - y^2}{x^4 + y^2}$ doesn't exist.

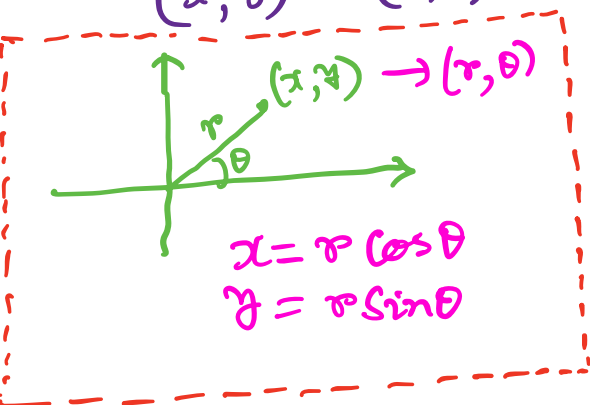
Two path Test (For non-existence of Limit):

If a function $f(x,y)$ has different limits along two different paths as (x,y) approaches (x_0, y_0) , then $\lim_{(x,y) \rightarrow (x_0, y_0)} f(x,y)$ doesn't exist.



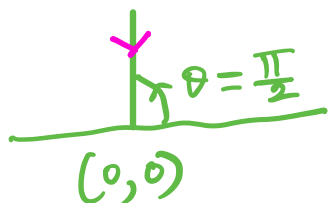
Ex(6): Using polar coordinates find

$$\lim_{(x,y) \rightarrow (0,0)} \frac{2x}{x^2+x+y^2} \quad \text{if it exists.}$$



$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \frac{2x}{x^2+x+y^2} &= \lim_{r \rightarrow 0} \frac{2r \cos \theta}{r^2 \cos^2 \theta + r \cos \theta + r^2 \sin^2 \theta} \\ &= \lim_{r \rightarrow 0} \frac{2r \cos \theta}{r^2 + r \cos \theta} \end{aligned}$$

$$= \lim_{r \rightarrow 0} \frac{2 \cos \theta}{r + \cos \theta} = \frac{2 \cos \theta}{\cos \theta} \quad \text{doesn't exist for } \theta = \frac{\pi}{2}$$



Hence $\lim_{(x,y) \rightarrow (0,0)} \frac{2x}{x^2+x+y^2}$ doesn't exist.

Ex(7): Find $\lim_{(x,y) \rightarrow (0,0)} \frac{y^2}{x^2+y^2}$ if it exists using polar coordinate.

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \frac{y^2}{x^2+y^2} &= \lim_{r \rightarrow 0} \frac{r^2 \sin^2 \theta}{r^2 \cos^2 \theta + r^2 \sin^2 \theta} \\ &= \lim_{r \rightarrow 0} \frac{r^2 \sin^2 \theta}{r^2} = \sin^2 \theta, \quad \text{different for different } \theta \end{aligned}$$

So, $\lim_{(x,y) \rightarrow (0,0)} \frac{y^2}{x^2+y^2}$ doesn't exist.

Note: (7) can also be done in Cartesian coordinate by approaching $(0,0)$ along the path $y = mx$ for different m .

Along the path $y = mx$,

$$\frac{y^2}{x^2+y^2} = \frac{m^2 x^2}{x^2 + m^2 x^2}$$
$$= \frac{m^2}{1+m^2}$$

So, $\lim_{\substack{(x,y) \rightarrow (0,0) \\ \text{along the path}}} \frac{y^2}{x^2+y^2} = \frac{m^2}{1+m^2}$ different for different m

Therefore, $\lim_{(x,y) \rightarrow (0,0)} \frac{y^2}{x^2+y^2}$ does not exist.

Properties of Limits for Function of Several Variables

Let f and g be functions from $\mathbb{R}^2$ to $\mathbb{R}$.

If $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L$ and $\lim_{(x,y) \rightarrow (x_0,y_0)} g(x,y) = M$

then (1) $\lim_{(x,y) \rightarrow (x_0,y_0)} [f(x,y) \pm g(x,y)] = L \pm M$

(2) $\lim_{(x,y) \rightarrow (x_0,y_0)} [f(x,y) \cdot g(x,y)] = L \cdot M$

(3) $\lim_{(x,y) \rightarrow (x_0,y_0)} [c f(x,y)] = cL$

(4) $\lim_{(x,y) \rightarrow (x_0,y_0)} \left[\frac{f(x,y)}{g(x,y)} \right] = \frac{L}{M}$ provided $M \neq 0$

(5) If r and s are integers with no common factors and $s \neq 0$,

$$\lim_{(x,y) \rightarrow (x_0,y_0)} [f(x,y)]^{r/s} = L^{r/s}$$

provided $L^{r/s}$ is a real number.

(e.g. if s is even, L can't be negative and we need to assume that $L > 0$)

- Limits of polynomials and rational functions at (x_0, y_0) can be calculated by evaluating at (x_0, y_0) , provided it is defined at (x_0, y_0)

Ex: Find $\lim_{(x,y) \rightarrow (0,0)} \frac{2x^2y}{x^4 + y^2}$

(Use two path test using paths $y = kx^2$ for different k)

Definition: Let $D \subseteq \mathbb{R}^2$

A function $f: D \rightarrow \mathbb{R}$ is said to be continuous at $(x_0, y_0) \in D$ if

(1) $(x_0, y_0) \in D$ (2) $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y)$ exists

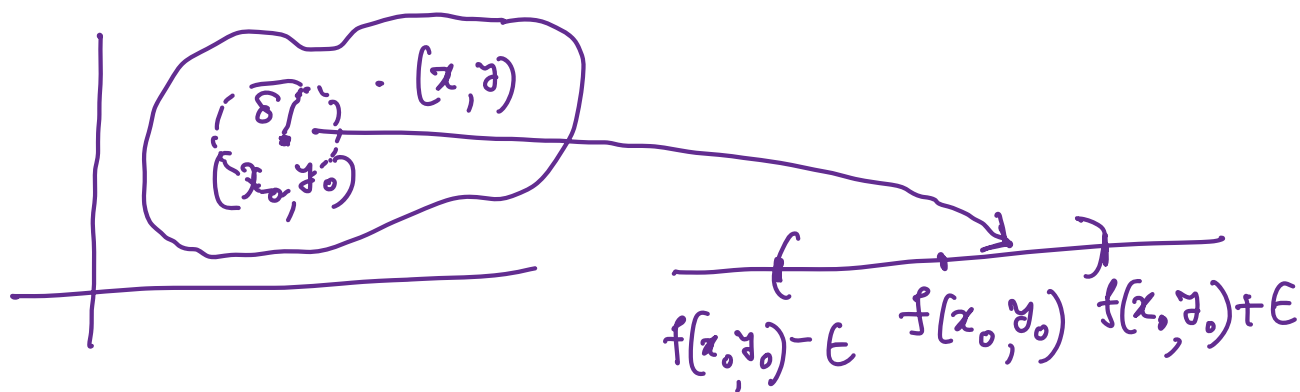
and (3) $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = f(x_0, y_0)$

Equivalently given any $\epsilon > 0$,
 $\exists \delta = \delta(\epsilon) > 0$ such that

(there exists)

$$\|(x,y) - (x_0, y_0)\| = \sqrt{(x-x_0)^2 + (y-y_0)^2} < \delta$$

$$\Rightarrow |f(x,y) - f(x_0, y_0)| < \epsilon$$



Note: f is called continuous if it is continuous at every point of its domain.

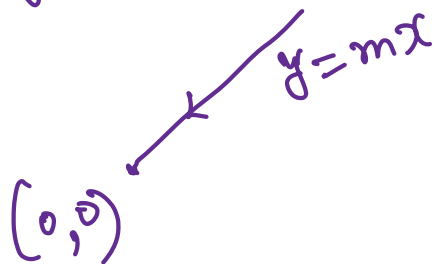
Ex: $f(x, y) = \frac{2xy}{x^2 + y^2}$ for $(x, y) \neq (0, 0)$
 $= 0$ for $(x, y) = (0, 0)$

Is f continuous at $(0, 0)$?

$f(0, 0) = 0.$

Question: Does $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ exist?

Along the line $y = mx$, $f(x, y) = \frac{2x \cdot mx}{x^2 + m^2 x^2}$



$= \frac{2m}{1+m^2}$

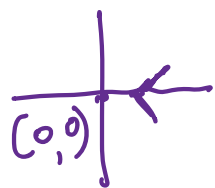
different for
different m

So, $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ does not exist
 and hence f is not continuous
 at $(0, 0)$.

Ex: Find $\lim_{(x, y) \rightarrow (0, 0)} \frac{3x^2 y}{x^2 + y^2}$ if it exists.

Note: Domain $(f) = \mathbb{R}^2 \setminus \{(0,0)\}$

check: $\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \frac{3x^2 \cdot 0}{x^2 + 0^2} = \frac{0}{x^2} = 0 \quad (x \neq 0)$



along $y=0$
($x \neq 0$)



$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \frac{3 \cdot 0^2 \cdot y}{0^2 + y^2} = \frac{0}{y^2} = 0 \quad (y \neq 0)$$

along $x=0$

($y \neq 0$)



But from here we can't conclude that $\lim_{(x,y) \rightarrow (0,0)} f(x,y)$ exists.

(0,0) However if the limit exists, it has to be zero.

Therefore, let us check using ϵ - δ definition if 0 is the limit.

Solution of the problem!

choose $\epsilon > 0$ arbitrarily.

$$\text{Now } |f(x,y) - 0| = \left| \frac{3x^2 y}{x^2 + y^2} - 0 \right|$$

$$= \frac{3x^2 |y|}{x^2 + y^2} \leq 3|y| \quad \left(\text{Since } \frac{x^2}{x^2 + y^2} \leq 1 \right)$$

$$\leq 3\sqrt{x^2 + y^2} < \epsilon$$

$$\text{if } \sqrt{(x-0)^2 + (y-0)^2} < \epsilon/3$$

Let $\delta = \frac{\epsilon}{3}$

Then $| (x-0)^2 + (y-0)^2 | < \delta \Rightarrow |f(x,y) - 0| < \epsilon$

So, $\boxed{\lim_{(x,y) \rightarrow (0,0)} f(x,y) = 0}$

Note: The same question can be asked in the following way.

Ex: Define $f(0,0)$ in such a way that extends f to be continuous at origin.

$$f(x,y) = \frac{3x^2y}{x^2+y^2}$$

Ex: Define $f(0,0)$ in such a way that extends f to be continuous at origin.

$$f(x,y) = \frac{4xy^2}{x^2+y^2}$$

Ex: Define $f(0,0)$ in such a way that extends f to be continuous at origin.

$$f(x,y) = \ln \left(\frac{3x^2 - x^2y^2 + 3y^2}{x^2+y^2} \right)$$

Let us find the limit using Polar coordinate $x = r \cos \theta$, $y = r \sin \theta$

$$\lim_{(x,y) \rightarrow (0,0)} f(x,y) = \lim_{r \rightarrow 0} \ln \left(\frac{3r^2 \cos^2 \theta - r^4 \cos^2 \theta \sin^2 \theta + 3r^2 \sin^2 \theta}{r^2 \cos^2 \theta + r^2 \sin^2 \theta} \right)$$

$$= \lim_{r \rightarrow 0} \ln \left(\frac{3r^2 - r^4 \cos^2 \theta \sin^2 \theta}{r^2} \right)$$

$$= \lim_{r \rightarrow 0} \ln(3 - r^2 \cos^2 \theta \sin^2 \theta) = \boxed{\ln 3}$$

Thus if we define $f(0,0) = \ln 3$,
then f will be continuous at $(0,0)$.

Ex: $f(x,y) = \frac{x^2}{x^2+y^2}$ for $(x,y) \neq (0,0)$
 $= 0$ for $(x,y) = (0,0)$.

Show that f is continuous at every point except $(0,0)$.

Ex: Given a function $f(x,y)$ and a positive number ϵ , show that there exists a $\delta > 0$ such that,
for all (x,y) with

$$\sqrt{x^2+y^2} < \delta \Rightarrow |f(x,y) - f(0,0)| < \epsilon$$

Given that $f(x,y) = \frac{x+y}{x^2+1}$ and $\epsilon = .01$

Note that $f(0,0) = 0$

$$\begin{aligned}\text{Now } |f(x,y) - f(0,0)| &= \left| \frac{x+y}{x^2+1} - 0 \right| \\ &= \frac{|x+y|}{x^2+1} \leq |x+y| \left(\text{since } x^2+1 \geq 1, \frac{1}{x^2+1} \leq 1 \right) \\ &\leq |x| + |y|\end{aligned}$$

$$\text{Now choose } \delta = \frac{\epsilon}{2} = \frac{.01}{2} = \boxed{.005}$$

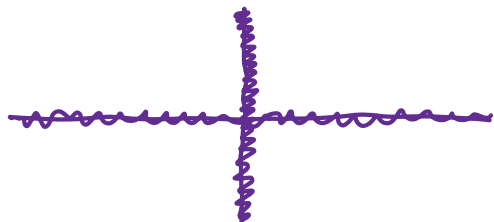
$$\begin{aligned}\text{Then } \sqrt{x^2+y^2} < .005 &\Rightarrow |x| < .005 \\ &\text{and } |y| < .005\end{aligned}$$

$$\begin{aligned}\Rightarrow |f(x,y) - f(0,0)| &\leq |x| + |y| \\ &< .005 + .005 = .01 = \epsilon\end{aligned}$$

Ex: At what points in the plane(space) are the following functions continuous.

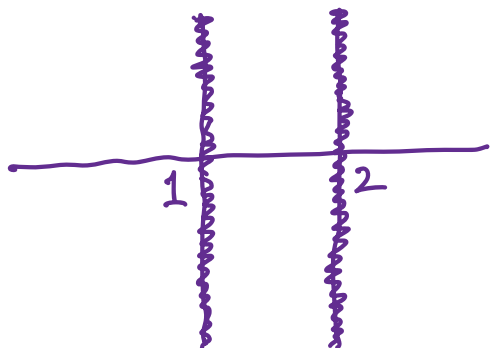
① $g(x,y) = \sin \frac{1}{xy}$

g is continuous for all (x,y) except $(x,0)$ or $(0,y)$



$$\textcircled{2} \quad h(x, y) = \frac{x^2 + y^2}{x^2 - 3x + 2}$$

h is continuous for all (x, y)
such that $x \neq 2$ and $x \neq 1$

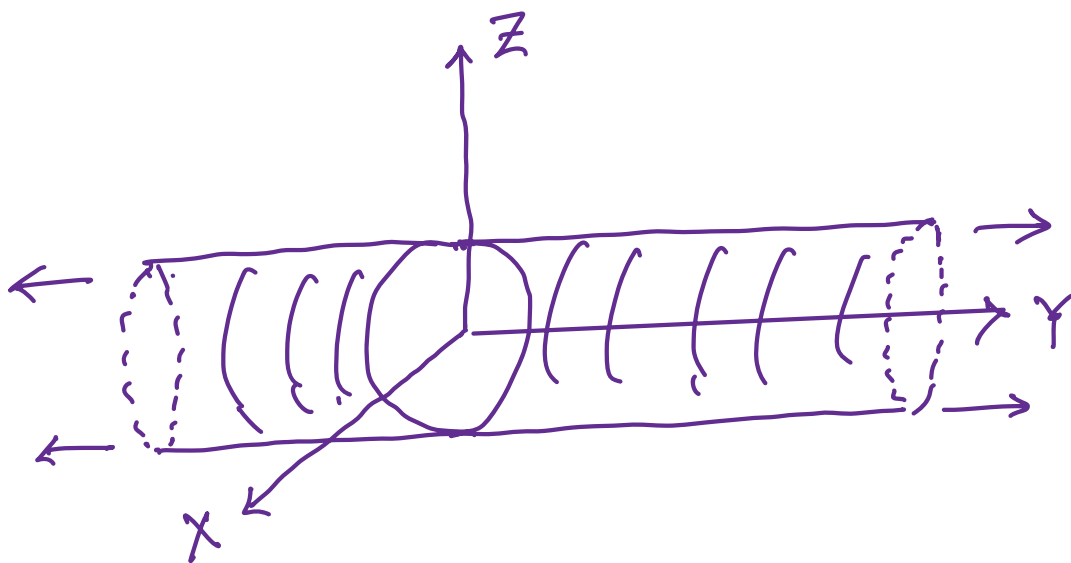


$$\textcircled{3} \quad g(x, y) = \frac{x + y}{2 + \cos x}$$

g is continuous for all (x, y)
($\cos x + 2$ is never zero)

$$\textcircled{4} \quad f(x, y, z) = \frac{1}{x^2 + z^2 - 1}$$

f is continuous for all (x, y, z)
s.t. $x^2 + z^2 \neq 1$



$$\textcircled{5} \quad h(x, y, z) = \frac{1}{|y| + |z|}$$

h is continuous for all (x, y, z)
s.t. $|y| + |z| \neq 0$ (i.e. if $(y, z) \neq (0, 0)$
(i.e. for all (x, y, z) except $(x, 0, 0)$)

